

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience using an AI coding assistant (Claude Code) to build a real end-to-end project, and a strong interest in extending this into evaluating and building AI agents and agentic workflows professionally. Combines a machine learning foundation (Scikit-Learn, Keras, TensorFlow) with practical data engineering experience at Deutsche Bundesbank, and comfort documenting and communicating technical work clearly.

Core Competencies

- **Applied AI Tools:** Hands-on use of an AI coding assistant (Claude Code) in a real project; strong interest in evaluating AI tools, models, and agentic workflows
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means
- **Programming:** Python, R, SQL, Java, C++
- **Data Engineering:** ETL/ELT pipelines, data quality automation, migration testing (Deutsche Bundesbank AnaCredit)
- **Documentation & Communication:** Structured reporting, stakeholder communication, MS Office (Word, Excel, PowerPoint)
- **Cloud & Tools:** Azure Cognitive Services, AWS, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to validate business logic after data migration
 - Built automated error detection identifying statistical deviations in large-scale reported banking data, with automated Jira ticketing and stakeholder communication

Projects

- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Designed and built an end-to-end data pipeline for simulated bank transactions (150 accounts), using Claude Code as an AI coding assistant to accelerate implementation
 - Built a normalized SQLite database with foreign-key integrity checks, staged data cleaning, and a fully auditable log recording every change
 - Produced quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering on 25+ years of national health data
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: machine learning & AI, data analysis & statistics, numerical optimization.
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.